

RISLAN MOHAMED

Associate Software Engineer · AI/ML Developer · Cybersecurity Researcher

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PROFESSIONAL SUMMARY

Results-driven Associate Software Engineer with hands-on experience in full-stack web development, AI/ML systems, database management, and cybersecurity. Skilled in building intelligent applications using TensorFlow, Claude AI, and Python, with a strong foundation in penetration testing and network infrastructure. Currently pursuing a Pearson BTEC Higher National Diploma in Computing (Software Engineering), combining academic research with real-world industry practice. Passionate about leveraging artificial intelligence to solve complex security and operational challenges.

TECHNICAL SKILLS

Programming Languages	Python · Java · PHP · JavaScript · HTML5 · CSS3
Web Development	Bootstrap · Responsive Design · PHP Backend · REST APIs · Cross-browser Compatibility
AI / Machine Learning	TensorFlow · Deep Learning · Streamlit · Scikit-Learn · NumPy · Pandas
Data Science & Viz	Matplotlib · Seaborn · Plotly · Jupyter Notebook · Statistical Analysis
Database Management	MySQL · phpMyAdmin · SQL · Relational Database Design & Optimization
Cybersecurity	Penetration Testing · Kali Linux · Metasploit · CVE Analysis · VLANs · Subnetting
Tools & DevOps	Git · GitHub · VS Code · Claude Code · Model Context Protocol (MCP)
Human Languages	English (Fluent) · Tamil (Native) · Sinhala (Conversational)

PROFESSIONAL EXPERIENCE

Associate Software Engineer <i>UNIX-NET-TECHNOLOGIES</i> - Build responsive web interfaces using HTML5, CSS3, Bootstrap, and JavaScript. - Develop secure PHP backend solutions and manage MySQL databases. - Handle web hosting, domain management, DNS configuration, and performance optimization.	November 2025 – Present
Website Manager <i>EuroFresh (seasonal online)</i> - Manage e-commerce platform — product listings, pricing, and inventory updates. - Maintain database accuracy to ensure reliable product data for customers.	May 2025 – Present
Freelance Software Engineer <i>Independent Consultant</i> - Deliver custom software, SQL databases, and network infrastructure for clients. - Design VLANs, subnetting, and security configurations; produce technical documentation.	March 2025 – Present
IT Administrator & Analyzer <i>Afran Travels & Tours</i> - Maintained systems and network infrastructure ensuring 99%+ uptime. - Enforced cybersecurity policies, managed user accounts, and provided technical support.	2023 – 2025

RESEARCH & PROJECTS

AI-Powered Automated Penetration Testing System

Nov 2025 – Jun 2026

Technologies: Python · Claude AI · MCP · Kali Linux · CVE RAG · DeepSeek-V3.2

- Built an AI-assisted penetration testing pipeline integrating Claude AI, MCP, and Kali Linux to automate vulnerability scanning, enrichment, and analysis.
- Developed a local CVE RAG engine enriching scan outputs with CVSS scores, severity ratings, and remediation recommendations.
- Evaluated using Precision, Recall, F1 Score, and Confusion Matrix; validated with DeepSeek-V3.2. (Final Thesis – BTEC HND, BCAS Campus)

AI-Powered Fabric Type Identifier for E-Commerce

April – July 2025

Technologies: Python · TensorFlow · Streamlit · Scikit-Learn · NumPy · Pandas · Matplotlib

- Developed a TensorFlow deep learning model to detect cotton fabric type and estimate cotton percentage with 95.7% accuracy.
- Integrated into a Streamlit app with real-time verification, confidence visualizations, downloadable reports, and watermarked outputs.
- Reduced customer return rates by providing AI-backed product verification on apparel e-commerce platforms.

TransitFlow Analytics System

Apr 2026 – Jun 2026

Technologies: Java 21 · Spring Boot · Python · Streamlit · H2 Database · REST API · Pandas · Plotly · JUnit 5 · PyTest · Selenium · MVC · SOLID Principles

- Developed a business intelligence platform capable of processing large-scale transportation and e-commerce datasets using memory-efficient techniques.
- Implemented analytics modules for revenue tracking, customer segmentation, demand forecasting, and logistics performance evaluation through interactive dashboards.
- Applied object-oriented design principles, design patterns, and automated testing frameworks to enhance system scalability, maintainability, and reliability.

EDUCATION

Pearson BTEC Higher National Diploma – Computing (Software Engineering)

November 2024 – September 2026

BCAS Campus – Kalmunai (Pearson College London)

Level 5 qualification covering software development, database systems, AI/ML, and cybersecurity. Served as Batch Representative and active member of BCAS IT Society (BITS).

GCE Advanced Level – Bio Science Stream

2020 – 2023

Addalaichchenai MMV National School

Completed A/Ls in Biology, Chemistry, and Physics; developed strong analytical and scientific reasoning skills.

CERTIFICATIONS

- **Anthropic** – Claude Code in Action (February 2026) (*Credential ID: 45g2nmewtct9*)
- **UniAthena** – Diploma in Data Visualization Using Python (*Score: 89%*)
- **ADBI E-Learning** – Cybersecurity: Principles and Data Security Tools
- **Simplilearn** – Introduction to AI · Cyber Security 101 · Digital Security Fundamentals · UI/UX Basics
- **Additional** – Cloud Computing · Machine Learning · Web Development

VOLUNTEERING & LEADERSHIP

- **Batch Representative** – BCAS IT Society (BITS), 2025–2026: Coordinated between students and faculty; helped organise tech events, workshops, and knowledge-sharing sessions.